

22CH203 – ENGINEERING CHEMISTRY II

UNIT V & NUCLEAR REACTIONS

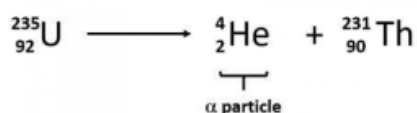
1. RADIOACTIVE DECAY (ALPHA, BETA AND GAMMA)

Radioactive decay is the emission of energy in the form of ionizing radiation. The ionizing radiation that is emitted can include alpha particles, beta particles and/or gamma rays. Radioactive decay occurs in unbalanced atoms called radionuclides.

Elements in the periodic table can take on several forms. Some of these forms are stable; other forms are unstable. Typically, the most stable form of an element is the most common in nature. However, all elements have an unstable form. Unstable forms emit ionizing radiation and are radioactive. There are some elements with no stable form that are always radioactive, such as uranium. Elements that emit ionizing radiation are called radionuclides.

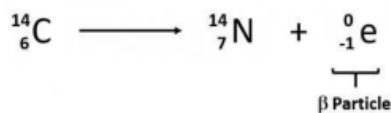
1.1. Alpha Particle (α)

Rutherford's experiments demonstrated that there are three main forms of radioactive emissions. The first is called an alpha particle, which is symbolized by the Greek letter α . An alpha particle is composed of two protons and two neutrons and is the same as a helium nucleus. It has a 2+ charge. When a radioactive atom emits an alpha particle, the original atom's atomic number decreases by two (because of the loss of two protons), and its mass number decreases by four (because of the loss of four nuclear particles). We can represent the emission of an alpha particle with a chemical equation— for example; the alpha-particle emission of uranium-235 is as follows:



1.2. Beta Particle (β)

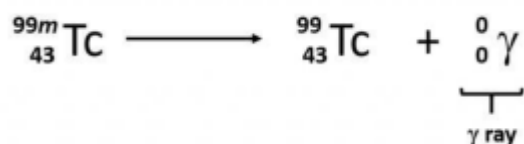
The second type of radioactive emission is called a beta particle, which is symbolized by the Greek letter β . A beta particle is an electron ejected from the nucleus (not from the shells of electrons about the nucleus) and has a -1 charge. We can also represent a beta particle as ${}^0_{-1}\text{e}$. The net effect of beta particle emission on a nucleus is that a neutron is converted to a proton. The overall mass number stays the same, but because the number of protons increases by one, the atomic number goes up by one. Carbon-14 decays by emitting a beta particle.



1.3. Gamma Radiation (γ)

The third major type of radioactive emission is not a particle but rather a very energetic form of electromagnetic radiation called gamma rays, symbolized by the Greek letter γ . Electromagnetic radiation can be characterized into different categories based on the wavelength and photon energies. Note that the human sensory adaptations of sight and hearing have evolved to detect electromagnetic radiation, with radio waves having wavelengths between 1 mm and 100 km and visible light having wavelengths between 380 – 700 nm. Technological advances have helped humankind utilize other forms of electromagnetic radiation including X-rays and microwaves.

Some electromagnetic radiation with very short wavelengths are active enough that they may knock out electrons out of atoms in a sample of matter and make it electrically charged. The types of radiation that can do this are termed ionizing radiation. X-rays and Gamma rays are examples of ionizing radiation. Some radioactive materials emit gamma radiation during their decay. For example, in the decay of radioactive technetium-99, a gamma ray is emitted. Note that in radioactive decay where the emission of gamma radiation occurs, that the identity of the parent material does not change, as no particles are physically emitted.



Alpha, beta, and gamma emissions have different abilities to penetrate matter. The relatively large alpha particle is easily stopped by matter (although it may impart a significant amount of energy to the matter it contacts). Beta particles penetrate slightly into matter, perhaps a few centimeters at most. Gamma rays can penetrate deeply into matter and can impart a large amount of energy into the surrounding matter. Table 1 summarizes the properties of the three main types of radioactive emissions and Figure 1 summarizes the ability of each radioactive type to penetrate matter.

Characteristic	Alpha Particles	Beta Particles	Gamma Rays
Symbols	α , ${}^4_2\text{He}$	β , ${}^0_{-1}\text{e}$	γ
Identity	Helium nucleus	electron	Electromagnetic radiation
Charge	2+	1-	None
Mass Number	4	0	0
Penetrating Power	Minimal (will not penetrate skin)	Short (will penetrate skin and some tissues slightly)	Deep (will penetrate tissues deeply)

Table 1 - The three main forms of Radioactive Emissions

Figure 1 is illustration of the relative abilities of three different types of ionizing radiation to penetrate solid matter. Typical alpha particles (α) are stopped by a sheet of paper, while beta particles (β) are stopped by an aluminum plate. Gamma radiation (γ) is damped when it penetrates lead.

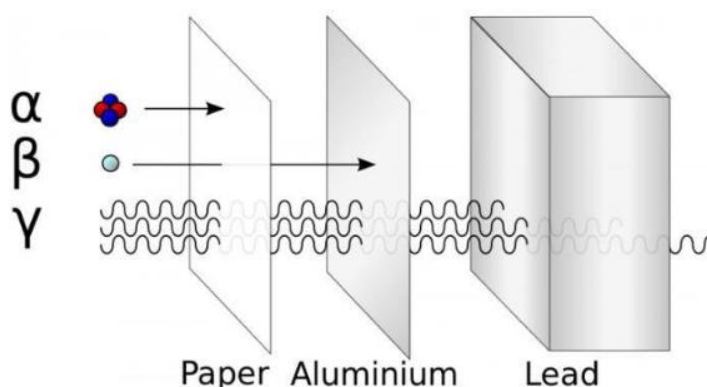


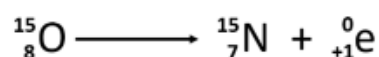
Figure 1 - Three different types of ionizing radiation to penetrate solid matter.

1.4. Other types of decay

1.4.1. Positron Emission (β^+ decay)

In addition to the three major types of radioactive particles listed above, two additional less common types of emissions have been discovered. These include positron emission and electron capture.

Positron emission (β decay) is the emission of a positron from the nucleus. Oxygen-15 is an example of a nuclide that undergoes positron emission:

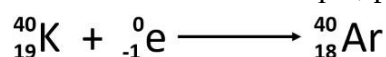


Positron emission is observed for nuclides in which the n:p ratio is low. These nuclides lie below the band of stability. Positron decay is the conversion of a proton into a neutron with the emission of a positron. The n:p ratio increases, and the daughter nuclide lies closer to the band of stability than did the parent nuclide. The positron has the mass of an electron, but a positive charge.

Thus, the overall mass of the nuclide doesn't change, but the atomic number is decreased by one, which causes a change in the elemental identity of the daughter isotope.

1.4.2. Electron capture

Electron capture occurs when one of the inner electrons in an atom is captured by the atom's nucleus. For example, potassium-40 undergoes electron capture.



Electron capture occurs when an inner shell electron combines with a proton and is

converted into a neutron. The loss of an inner shell electron leaves a vacancy that will be filled by one of the outer electrons. As the outer electron drops into the vacancy, it will emit energy. In most cases, the energy emitted will be in the form of an X-ray. Like positron emission, electron capture occurs for “proton-rich” nuclei that lie below the band of stability. Electron capture has the same effect on the nucleus as positron emission: The atomic number is decreased by one and the mass number does not change. This increases the n:p ratio, and the daughter nuclide lies closer to the band of stability than did the parent nuclide. Whether electron capture or positron emission occurs is difficult to predict. The choice is primarily due to kinetic factors, with the one requiring the smaller activation energy being the one more likely to occur.

Figure 2 summarizes these types of decay, along with their equations and changes in atomic and mass numbers.

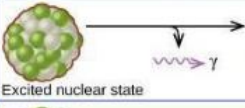
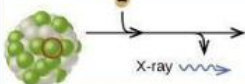
Type	Nuclear equation	Representation	Change in mass/atomic numbers
Alpha decay	${}^A_ZX \rightarrow {}^4_2\text{He} + {}^{A-4}_{Z-2}Y$		A: decrease by 4 Z: decrease by 2
Beta decay	${}^A_ZX \rightarrow {}^0_{-1}e + {}^{A}_{Z+1}Y$		A: unchanged Z: increase by 1
Gamma decay	${}^A_ZX \rightarrow {}^0_0\gamma + {}^A_ZY$		A: unchanged Z: unchanged
Positron emission	${}^A_ZX \rightarrow {}^0_{+1}e + {}^{A}_{Z-1}Y$		A: unchanged Z: decrease by 1
Electron capture	${}^A_ZX + {}^0_{-1}e \rightarrow {}^{A}_{Z-1}Y + \gamma$		A: unchanged Z: decrease by 1

Figure 2 - Summary of the type, nuclear equation, representation, and any changes in the mass or atomic numbers for various types of decay.

DISCUSSION QUESTIONS:

1. Identify the three types of radioactive decays of unstable isotopes
2. Compare the penetrating power of the alpha, beta and gamma.
3. Differentiate positron emission of a radioactive nucleus for electron capture reaction.